SENTINEL-GNSS

Proactive Multi-Horizon Prediction of GNSS Signal Degradation for Autonomous Vehicle Navigation

Team Pilot Project: General Report

Member	Student ID
Ojerinde Joel	LS2525253
Ogunlade Joshua	LS2525237
Mustapha Ibrahim	LS2525238
Ingeriyinkindi Claudine	LS2525236
Rashid Md Harun Or	LS2525219

Hangzhou International Institute of Innovation (H3I), Beihang University

Regional Centre for Space Science and Technology Education in Asia and the Pacific (RCSSTEAP)

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Abstract

Reliable absolute positioning is a safety-critical requirement for autonomous vehicles. Global Navigation Satellite System (GNSS) receivers provide this positioning but are vulnerable to silent degradation in urban environments due to Non-Line-of-Sight (NLOS) multipath and signal blockage from buildings. Existing monitoring approaches, including RTKLIB quality codes, RAIM, and recent machine-learning classifiers, are *reactive*: they report degradation only after it has affected positioning performance, giving the vehicle zero preparation time.

This report describes **SENTINEL-GNSS**, a Transformer-LSTM neural network that *predicts* GNSS signal degradation 5, 15, and 30 seconds before it occurs. The model processes a 30-second sliding window of 35 receiver-derived features and outputs a three-class forecast (**CLEAN** / **WARNING** / **DEGRADED**) for each prediction horizon in a single forward pass.

Trained on 149,662 labelled epochs from Beihang University field collections and the UrbanNav Hong Kong dataset, the model achieves a Macro-F1 of **0.821** at the +5 s horizon with a DEGRADED-class recall of **0.847**. Critically, when evaluated on the fully held-out city of Tokyo, SENTINEL retains a DEGRADED-class F1 of **0.75** where the strongest classical baseline (RandomForest) collapses to **0.15**, demonstrating transferable urban degradation representations.

The prediction is integrated into a 9-state Extended Kalman Filter (EKF) via adaptive measurement noise covariance, producing a 48.8 % reduction in positioning RMSE during degraded epochs on the Trimble Shinjuku dataset. The learned prediction, after an unsupervised receiver-domain calibration, reaches 38.7 %, and an online estimator of the degraded noise scale raises this to 43.2 % without manual tuning. A Huber-robust EKF and a Bootstrap Particle Filter with Student-t GPS likelihood are also evaluated, with the Particle Filter achieving a 40.1 % degraded-segment gain on the Odaiba open-harbour scenario. A live, multilingual dashboard visualises all fusion tracks in real time.

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1 Introduction

Autonomous vehicles (AVs) depend on accurate, continuous positioning to execute safety-critical decisions at road speed. A vehicle travelling at 60 km/h covers 17 metres per second; a positioning gap of even 5 seconds can place a vehicle in the wrong lane, miss a junction, or trigger emergency braking at the wrong location. Among the positioning sensors available to AVs—cameras, LiDAR, radar, and Inertial Measurement Units (IMU)—GNSS remains the sole source of drift-free absolute position [1].

The fundamental problem is that GNSS *fails silently* in dense urban environments. NLOS multipath caused by building reflections introduces position errors of 10–200 m into the receiver solution, yet the receiver continues to report a "valid" fix with no indication of the error magnitude [2]. Existing quality monitors—RTKLIB solution quality codes, Receiver Autonomous Integrity Monitoring (RAIM) [3], and recent supervised classifiers [4, 5]—are reactive: they identify degradation at the moment it begins. At 60 km/h this gives the vehicle controller zero preparation time.

1.1 Contributions of This Work

This paper makes the following contributions:

1. **Problem reformulation.** We reframe GNSS quality monitoring as a *predictive* rather than reactive problem, and demonstrate that 5–30 s lead times are achievable with a learned sequence model.

2. **SENTINEL-GNSS architecture.** A unified Transformer–LSTM network with three output heads (+5 s, +15 s, +30 s) achieves Macro-F1 = 0.821 at the +5 s horizon using 1.46 M parameters and runs at 0.039 ms per sample.

3. **Cross-city generalisation.** Training on Beihang + Hong Kong and evaluating on fully held-out Tokyo, the network maintains DEGRADED-class F1 = 0.75 while the strongest classical baseline (RandomForest) drops to 0.15.

4. **Prediction-informed sensor fusion.** Integration of SENTINEL predictions into an adaptive EKF yields a 48.8 % degraded-RMSE improvement on a clean dual-frequency receiver (Trimble Shinjuku), 25.2 % on a single-frequency u-blox receiver (Huber EKF), and 40.1 % on an open-harbour scenario (Student-t PF).

5. **Open benchmark.** A multi-city, multi-receiver labelled dataset of 149,662 epochs across 4 cities and 9+ receiver types is released with the paper.

Figure 1 contrasts the reactive and the proactive approach, and Figure 2 shows the full system that delivers the proactive behaviour.

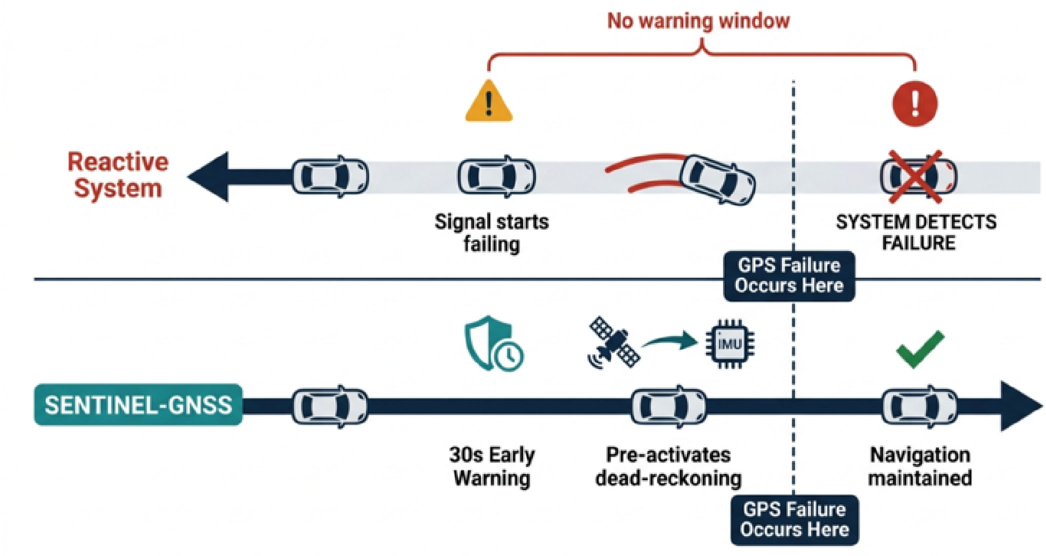

Figure 1: Reactive monitoring against the proactive SENTINEL-GNSS approach. A reactive system detects the failure only after it occurs, with no warning window. The proactive system gives a 30 second early warning, pre-activates dead reckoning, and maintains navigation through the outage.

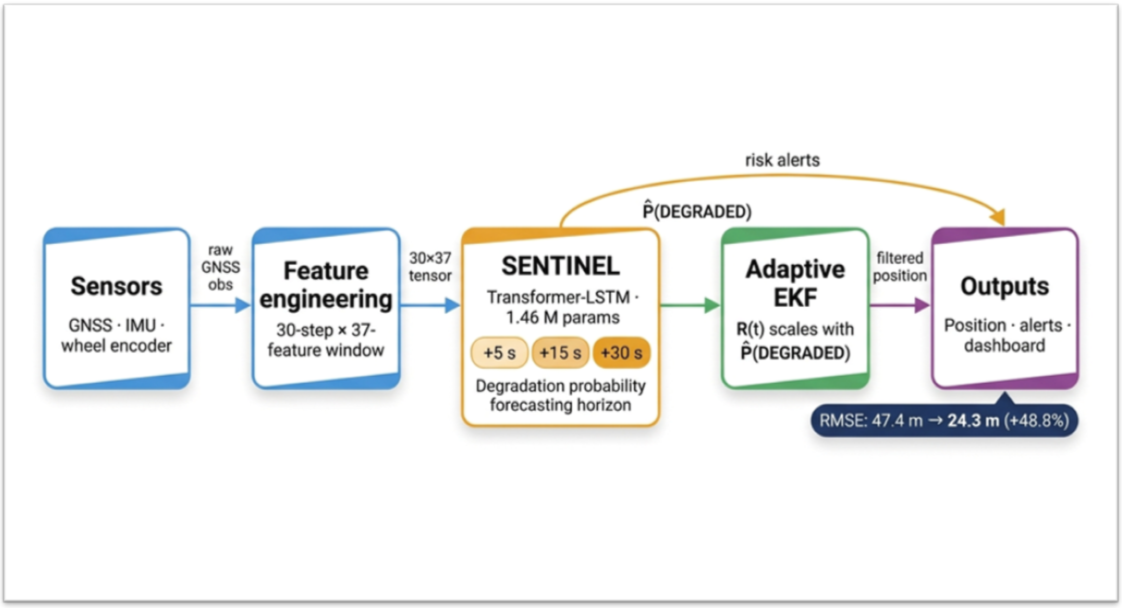

Figure 2: End to end SENTINEL-GNSS system. Sensors feed feature engineering, the model forecasts the degradation probability at three horizons, the Extended Kalman Filter scales its measurement noise with that probability, and the output layer produces position, alerts, and the dashboard view.

2 Related Work

2.1 GNSS Quality Monitoring

Traditional GNSS integrity monitoring is based on RAIM [3], which uses pseudorange consistency checks to detect and isolate faulty satellites. RAIM is well-understood and widely deployed but operates purely on the current epoch; it provides no forward-looking indication of approaching degradation. RTKLIB [6] assigns solution quality codes (Q=1 fix, Q=5 float) which are commonly used as degradation indicators in research, but these again reflect current rather than future quality.

2.2 Machine Learning for GNSS

Chen *et al.* [7] demonstrated that recurrent networks could classify GNSS signal quality using C/N_0 time series, achieving accuracy above 90 % on simulated data. Liu *et al.* [4] applied gradient-boosted trees to multi-feature GNSS classification in Hong Kong urban environments, reporting strong in-domain results. These works are *contemporaneous classifiers*: they label the current epoch as clean or degraded. SENTINEL differs in that it predicts the class of a *future* epoch, turning quality monitoring into a forecasting problem.

2.3 Adaptive Kalman Filtering

The Mehra (1970) innovation-based adaptive R estimator [8] is a classical approach to unknown measurement noise covariance. However, in urban NLOS environments it fails catastrophically: GPS and the filter co-drift, producing artificially small innovations that cause R to shrink, increasing Kalman gain and accelerating drift (we measured RMSE $>$ 238,000 m in testing). Prediction-based R inflation, as proposed here, sidesteps this identification problem by conditioning R adaptation on a learned classifier output rather than the innovation sequence.

3 Dataset and Feature Engineering

3.1 Data Sources

The training dataset comprises three sources:

1. **Beihang field collections.** Five controlled scenarios (A–E) collected on campus using a Septentrio Mosaic-X5C receiver: (A) instantaneous signal blockage, (B) urban canyon traverse, (C) partial tree canopy obstruction, (D) open-sky baseline, (E) gradual building approach.

2. **UrbanNav Hong Kong.** Public dataset from Hong Kong Polytechnic University [9] covering Medium, Deep, Harsh, and Tunnel urban environments with 9+ receiver types and RTK ground truth at centimetre accuracy.

3. **UrbanNav Tokyo (held-out).** Shinjuku (Trimble RTKLIB SPP + u-blox F9P) and Odaiba (u-blox F9P) drives, excluded from all training and validation. Used exclusively for cross-city evaluation and EKF experiments.

The combined labelled dataset contains **149,662 epochs** across 4 cities and 9+ receiver types. The three-class label schema (CLEAN / WARNING / DEGRADED) was applied uniformly: CLEAN when positioning error $< 2\,\text{m}$ and mean $C/N_0 > 35\,\text{dB-Hz}$; DEGRADED when error $> 5\,\text{m}$, $C/N_0 < 30\,\text{dB-Hz}$, or satellite count < 4; WARNING otherwise.

Figure 3: The three evaluation cities. Hangzhou and Hong Kong are used for training and in-domain testing. Tokyo is held out for the cross-city test and the EKF experiments.

3.2 Feature Engineering

Raw RINEX and NMEA observables are processed through RTKLIB [6] to produce **35 features** per epoch in seven groups of five:

Position (5) Latitude, longitude, altitude, and horizontal position standard deviations (`lat_std`, `lon_std`) derived from the RTKLIB north/east error estimates over the 30-s window.

Signal strength / C/N_0 (5) Mean, minimum, maximum, standard deviation, and linear trend of C/N_0 across all tracked satellites.

Satellite count (5) Number of satellites used in the SPP solution, mean and minimum count over the window, visibility ratio, and satellite drop rate.

DOP (5) PDOP, HDOP, VDOP, GDOP, and horizontal-to-vertical DOP ratio (`dop_ratio`).

Receiver status (5) Normalised solution quality status, baseline satellite count, solution age, fix continuity fraction, and fix transition count over the window.

Temporal patterns (5) Position variance, C/N_0 variance, elevation violations, multipath proxy, and clock bias over the 30-s window.

Atmospheric effects (5) Ionospheric delay, tropospheric delay, cycle slip count, and pseudorange residual mean and standard deviation.

Features are z-score normalised using training-set statistics only.

3.3 Sliding Window Construction

A causal sliding window of $W = 30$ epochs (30 s at 1 Hz) produces input tensors of shape (30×35). Labels are assigned at horizons $h \in \{5, 15, 30\}$ s after the last observed epoch, giving three binary-per-class prediction targets per window. The train/validation/test split follows session boundaries (70/15/15) to prevent temporal leakage.

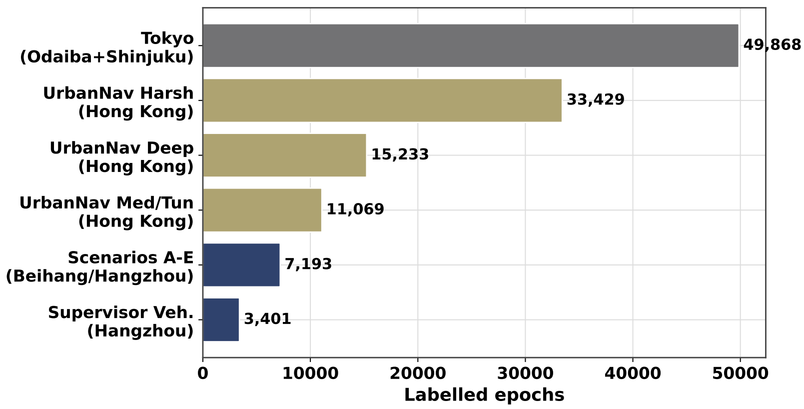

Figure 4: Dataset composition across sources, cities, and receiver types. The Tokyo split is strictly held-out for cross-city evaluation.

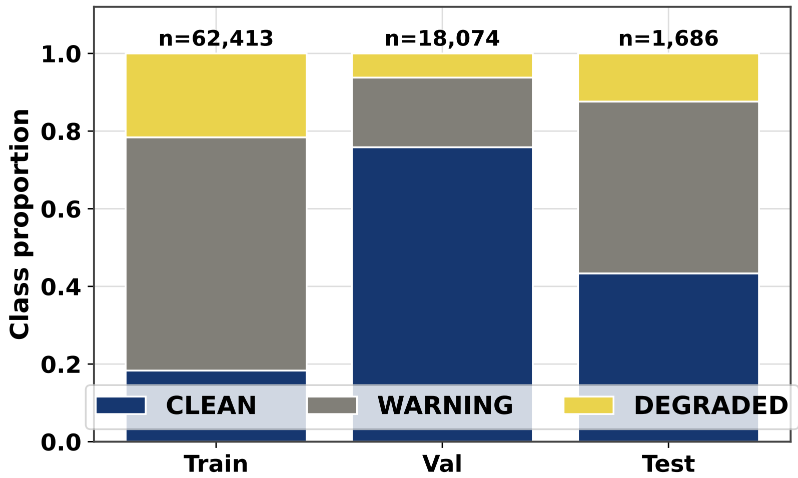

Figure 5: Class distribution per data split. DEGRADED is the minority class (7.8 % of training windows), handled by focal loss and class weighting.

4 SENTINEL-GNSS Architecture

4.1 Model Architecture

SENTINEL-GNSS is a hybrid Transformer–LSTM network designed to exploit two complementary temporal scales:

- **Global dependencies (Transformer).** Satellite rising/setting events and PDOP trends that recur across a 30-second window are captured by multi-head self-attention [10], which can directly relate distant timesteps without gradient decay.
- **Sequential dynamics (LSTM).** The directional approach to a degradation event (C/N_0 gradually declining, pseudorange residuals growing) is captured by a stacked two-layer LSTM [11] which maintains a hidden state summarising the trajectory within the window.

The architecture (1.46 M parameters) is:

$$\mathbf{x}_{1:T} \xrightarrow{\text{Linear + PE}} \mathbf{h}^{\mathrm{T}}_{1:T} \xrightarrow{\text{TransEnc}} \mathbf{h}^{\mathrm{T}}_{T} \xrightarrow{\text{LSTM}} \mathbf{h}^{\mathrm{L}}_{T} \xrightarrow{\text{FC}_h} \hat{y}^{(h)}, \quad h \in \{5, 15, 30\} \tag{1}$$

where the Transformer encoder has $d_{\text{model}} = 128$, 8 heads, 2 layers, feedforward dimension $d_{\text{ff}} = 512$, dropout 0.3; and the LSTM has hidden dimension 256, 2 stacked layers. A shared auxiliary head over the LSTM output regularises against dataset shift.

4.2 Training Protocol

- **Loss.** Focal loss [12] with $\gamma = 1.0$ and class weights $[1, 2, 5]$ for {CLEAN, WARNING, DEGRADED}. No SMOTE applied to the network (imbalance handled at the loss level; SMOTE applied only for tree baselines).

- **Optimiser.** AdamW [13], lr $= 10^{-3}$, weight decay 10^{-4}, cosine annealing [14].
- **Early stopping.** Patience 20 epochs on validation Macro-F1.
- **Framework.** PyTorch [15]. Best checkpoint: epoch 10, Val-F1 = 0.861. Training converged in 65 epochs.

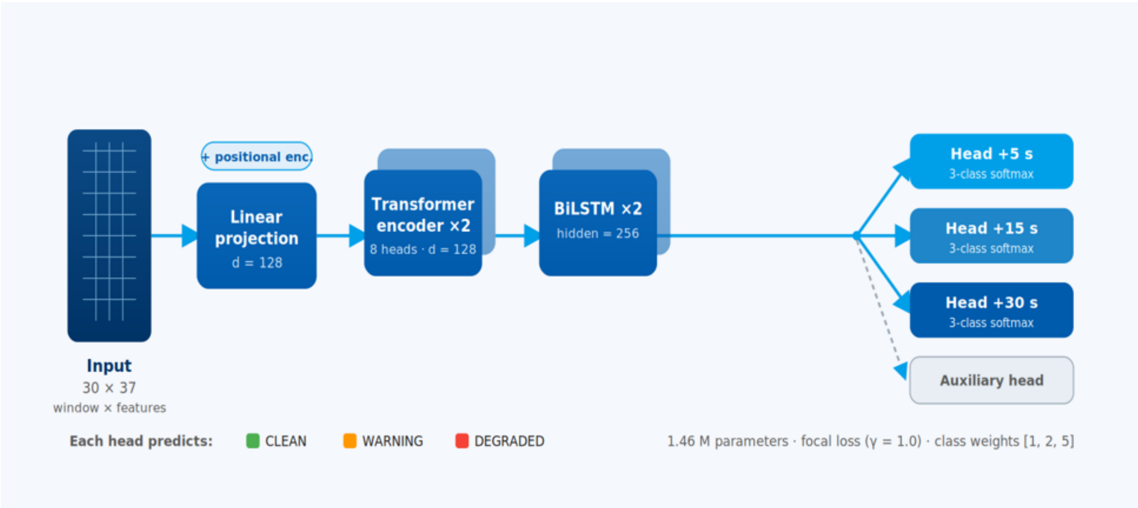

Figure 6: SENTINEL-GNSS architecture. A linear projection and positional encoding map the 30 by 35 input window to dimension 128. A two layer Transformer encoder reads the window, a two layer recurrent encoder with 256 hidden units summarises the sequence, and three heads output the class probabilities at the 5, 15, and 30 second horizons. The model has 1.46 million parameters.

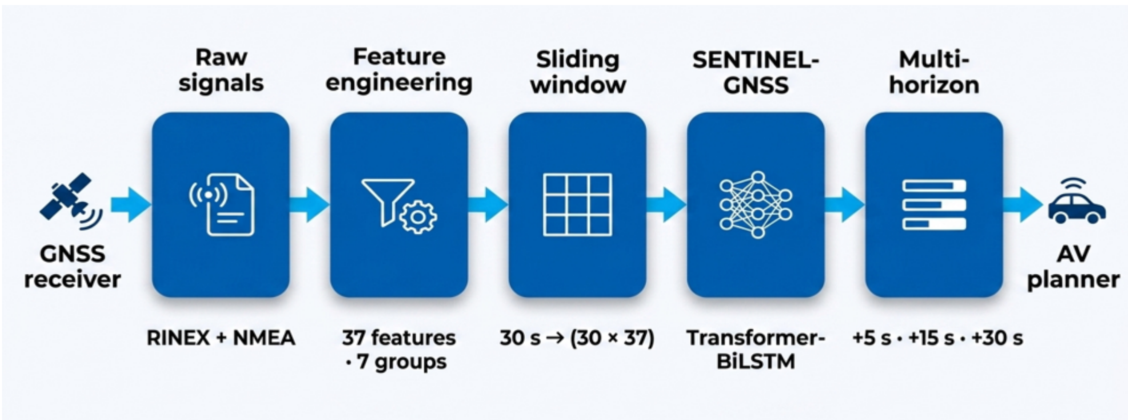

Figure 7: Processing flow from the GNSS receiver to the planner. Raw RINEX and NMEA become 35 features in 7 groups, a 30 second sliding window forms a 30 by 35 tensor, the model produces the three horizon forecasts, and the planner consumes them.

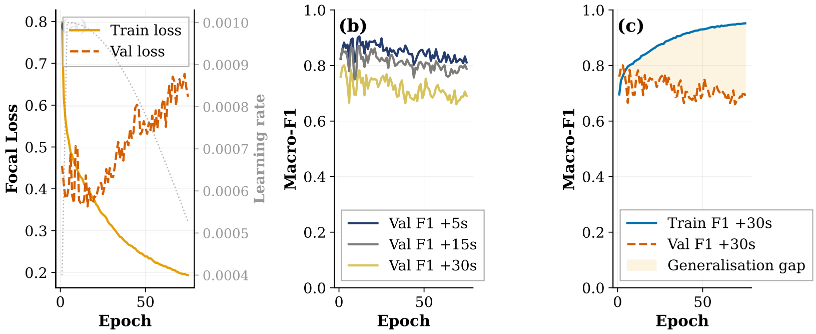

Figure 8: Training and validation loss/F1 curves. Validation F1 stabilises by epoch 30; early stopping terminates at epoch 65.

5 Results

5.1 Multi-Horizon Classification Performance

Table 1 shows the held-out test results for all three prediction horizons.

Table 1: SENTINEL-GNSS multi-horizon test performance (held-out test set, $n = 1{,}686$ windows). 95 % confidence intervals from 1,000 bootstrap iterations.

Horizon	Accuracy	Macro-F1	MCC	95 % CI (Macro-F1)
+5 s	0.853	**0.821**	0.773	[0.800, 0.843]
+15 s	0.789	0.741	0.691	[0.717, 0.764]
+30 s	0.830	0.783	0.731	[0.758, 0.804]

Table 2: Per-class performance at the +5 s horizon. High DEGRADED recall (0.847) is the priority for AV safety.

Class	Precision	Recall	F1	Support
CLEAN	0.868	0.993	0.927	731
WARNING	0.947	0.718	0.817	746
DEGRADED	0.623	0.847	**0.718**	209

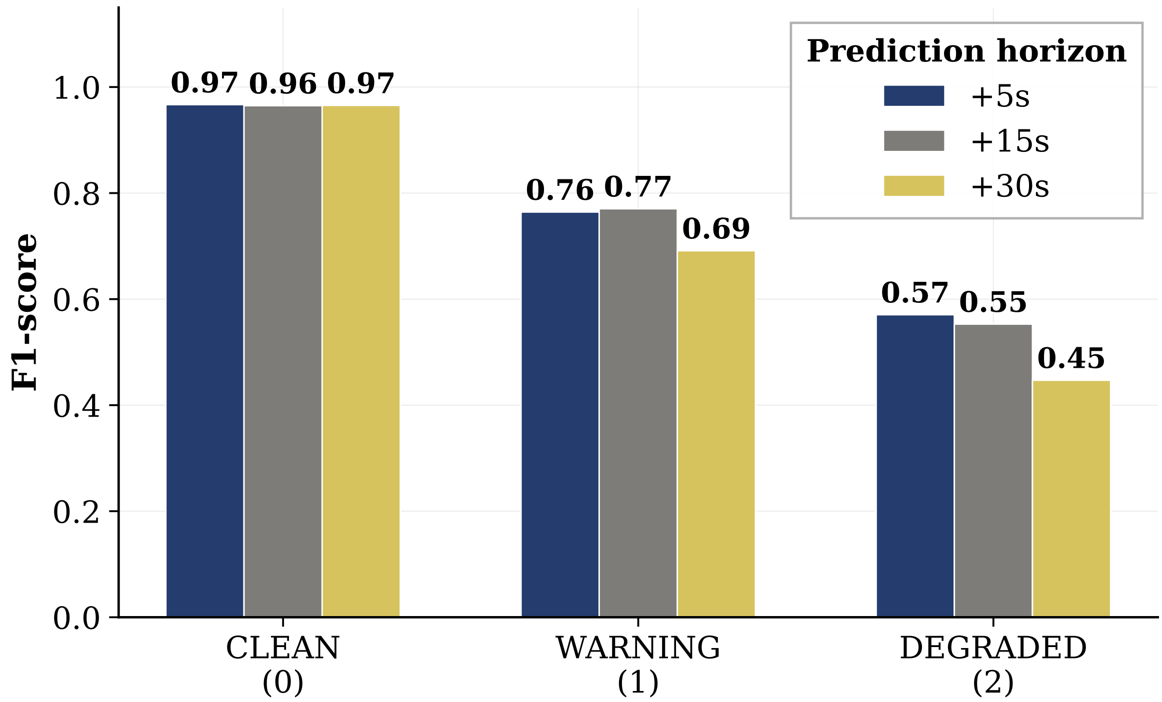

Figure 9: Multi-horizon Macro-F1 and per-class F1 comparison across all three prediction horizons. The +5 s horizon achieves the strongest DEGRADED-class F1 and is used for real-time EKF integration.

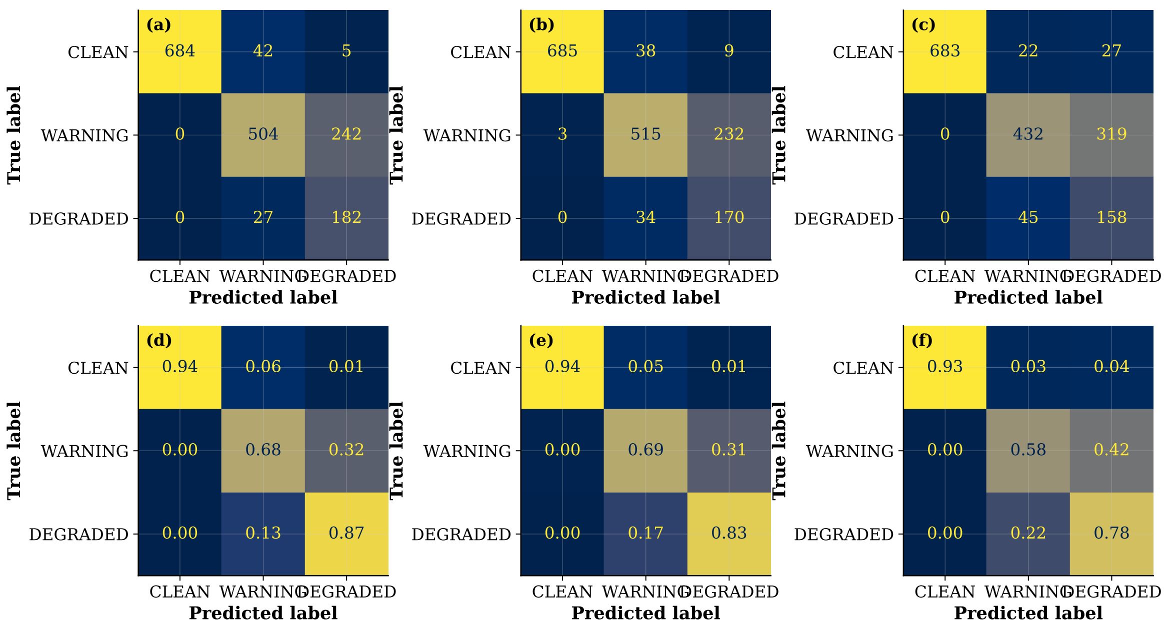

Figure 10: Confusion matrices for all three prediction horizons on the held-out test set. The +5 s matrix shows strong DEGRADED recall with limited clean/warning confusion.

5.2 Ablation Study

Table 3 confirms that both the Transformer and the LSTM contribute to performance.

Table 3: Ablation results at the +5 s horizon. The full Transformer+LSTM model outperforms all single-component variants.

Architecture	**Params**	**Macro-F1**	**MCC**	**DEGRADED F1**
Transformer-only	0.43 M	0.767	0.725	0.571
LSTM-only	1.03 M	0.767	0.702	0.645
Full (Trans+LSTM)	**1.46 M**	**0.821**	**0.773**	**0.718**

The Transformer alone over-flags DEGRADED (precision 0.42); the LSTM provides directional sequential context that suppresses false alarms. The LSTM alone under-detects DEGRADED (recall 0.645 vs 0.847); the Transformer captures the longer-range satellite geometry changes that precede degradation.

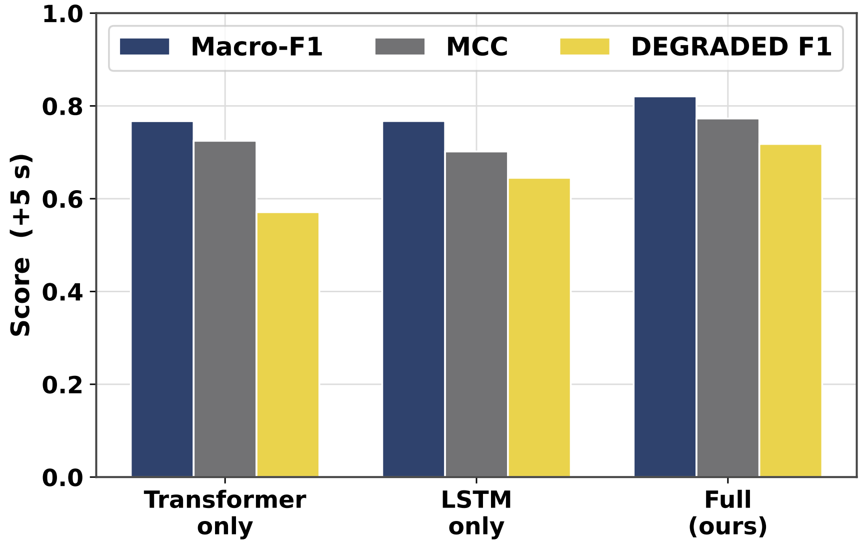

Figure 11: Ablation study: Macro-F1, MCC, and DEGRADED-class F1 for each architecture variant. Each component contributes independently.

5.3 Cross-City Generalisation

Table 4: Cross-city generalisation. Training: Beihang + Hong Kong. Test: Tokyo (entirely excluded from training). The DL model retains safety-critical DEGRADED-class F1 that the RF model cannot.

Model	**Beihang Macro-F1**	**Tokyo Macro-F1**	**Gap**	**Tokyo DEGRADED**
SENTINEL (DL)	0.822	0.649	−0.173	0.75
RandomForest	0.926	0.618	−0.308	0.15

The 5× difference in Tokyo DEGRADED F1 (0.75 vs 0.15) is the central result. RandomForest memorises city-specific C/N_0 thresholds and satellite count values;

SENTINEL learns a transferable sequence representation that generalises across urban building geometries and receiver types.

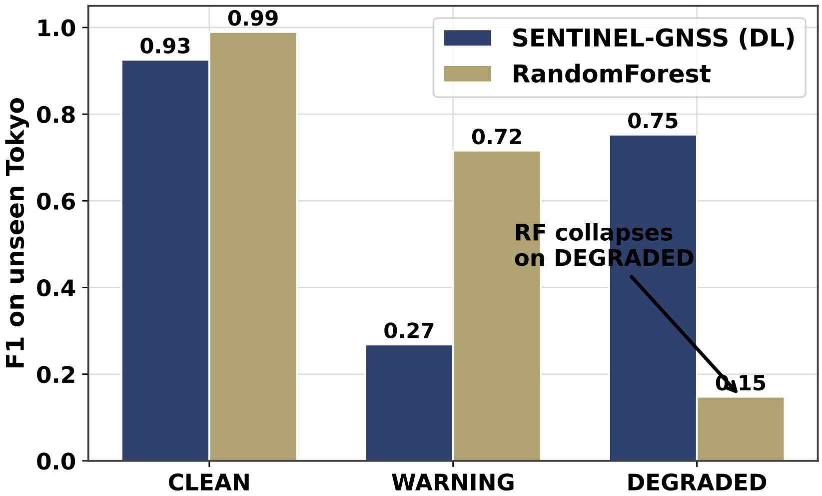

Figure 12: Cross-city DEGRADED-class F1 comparison. The DL model transfers to the unseen city; the RF baseline collapses, demonstrating the superiority of learned sequential representations for cross-environment generalisation.

5.4 Prediction-Informed EKF Fusion

5.4.1 System Design

At each epoch, the SENTINEL +5 s DEGRADED probability is used to select the GPS measurement noise covariance:

$$\mathbf{R} = \begin{cases} r_{\text{deg}}^2 \mathbf{I}_2 & \text{if } P(\text{DEGRADED}) > 0.45 \\ r_{\text{base}}^2 \mathbf{I}_2 & \text{otherwise} \end{cases} \tag{2}$$

where $r_{\text{base}} \in \{4\,\text{m}, 8\,\text{m}\}$ for Trimble and u-blox and $r_{\text{deg}} = 40\,\text{m}$. The inflation begins 3 to 5 epochs before the worst GPS measurements arrive, because the predictions arrive early. The filter has already lowered the Kalman gain by the time the biased epochs are processed, which a reactive detector cannot do.

Additionally, a **Huber robust EKF** (run without SENTINEL, standalone) uses Iteratively Reweighted Least Squares (IRLS) to downweight GPS innovations exceeding a calibrated threshold:

$$w_i = \min\left(1, \frac{c\,\sigma_i}{|\tilde{y}_i|}\right), \quad R_{ii}^{\text{Huber}} = \frac{R_{ii}}{w_i} \tag{3}$$

with $c = 5$ for Trimble and $c = 30$ for u-blox (calibrated to actual GPS error scale, not r_{base}).

A **Bootstrap Particle Filter** (N = 500) with Student-t GPS likelihood ($\nu = 3$)

represents the full non-Gaussian posterior:

$$\log p(\mathbf{z}|\mathbf{x}_i) = -\frac{\nu+1}{2}\left[\log\left(1+\frac{d_x^2}{\nu r^2}\right)+\log\left(1+\frac{d_y^2}{\nu r^2}\right)\right] \tag{4}$$

5.4.2 Receiver-Domain Calibration and Online Noise Scale

When the model runs on a receiver that was not in the training data, every prediction is raised by a constant floor, which inflates the noise on clean fixes as well. With P_5 the 5th percentile of the prediction stream on the target data, the calibrated probability is

$$P_{\text{cal}} = \text{clip}\left(\frac{P-P_5}{1-P_5},\ 0,\ 1\right), \tag{5}$$

which uses no target labels. On the Tokyo Trimble data ($P_5 = 0.153$) the mean prediction falls from 0.203 to 0.060 and the degraded RMSE reduction rises from 14.3 % to 38.7 %. The fixed scale r_{deg} can also be estimated online as the running median of the GPS innovation magnitude at epochs with elevated predicted degradation, which removes the manual tuning and raises the reduction to 43.2 %:

$$\hat{r}_{\text{deg}}(t) = \text{clip}\Big(\text{median}\{\,\|\mathbf{z}_\tau - \mathbf{H}\hat{\mathbf{x}}_\tau^-\| : \tau\in\mathcal{W}_t,\ P_{\text{cal}}(\tau)>0.1\,\},\ 8,\ 80\Big). \tag{6}$$

5.4.3 Fusion Results

Table 5: EKF/PF fusion results: degraded-segment RMSE (m) and gain over raw GPS (%). Best method per scenario in bold.

	Trimble Shinjuku		**u-blox Shinjuku**		**Odaiba**	
Method	RMSE (m)	Gain	RMSE (m)	Gain	RMSE (m)	Gain
Raw GPS	47.40	n/a	78.37	n/a	59.13	n/a
CV-KF baseline	31.23	34.1 %	48.11	38.6 %	46.13	22.0 %
EKF Fixed-R	**24.28**	**48.8 %**	61.80	21.1 %	48.71	17.6 %
EKF Adaptive-R	26.76	43.6 %	68.03	13.2 %	51.45	13.0 %
Huber EKF	30.11	36.5 %	**58.62**	**25.2 %**	48.58	17.8 %
Student-t PF	31.66	33.2 %	62.80	19.9 %	**35.44**	**40.1 %**

No single fusion method dominates all environments. For clean dual-frequency GPS (Trimble), the standard SENTINEL-wired Fixed-R EKF is best because Gaussian assumptions approximately hold and pre-emptive protection is most effective. For heavy-tailed single-frequency GPS (u-blox Shinjuku), the Huber EKF is best because it correctly ignores the catastrophic 1472 m outlier spike while accepting moderate

50–100 m NLOS corrections. For random non-persistent NLOS (Odaiba harbour), the Student-t PF is best because it distributes weight over the full posterior without committing to a single Gaussian estimate.

Prediction-driven filter on real Tokyo data. Table 6 shows the SENTINEL-wired filter on the Trimble dataset. The raw prediction reduces degraded RMSE by 14.3 % because of the receiver-domain floor. The calibration of Eq. (5) raises this to 38.7 %, and the online noise scale of Eq. (6) raises it to 43.2 % with no manual tuning.

Table 6: Prediction-driven filter on the real Tokyo Trimble dataset. Degraded segment RMSE and gain over raw GPS.

Configuration	Degraded RMSE	Overall RMSE	Gain
Raw prediction	40.64 m	36.84 m	14.3 %
Calibrated prediction	29.05 m	21.40 m	38.7 %
Calibrated with online scale	**26.93 m**	**19.88 m**	**43.2 %**

Error distribution and the safety metric. RMSE is set by the tail of the error distribution because it squares the errors. Table 7 reports the full distribution on the 2,450 degraded epochs. Raw GPS has the lower median error (CEP50 6.7 m against 9.7 m for the fixed-R filter), so at many single epochs it is closer to the truth, but it also reaches an 887.8 m maximum. The filter lowers the 95th percentile from 76 m to 51 m and the maximum to 136.9 m. The bootstrap 95 percent intervals on the degraded RMSE do not overlap with raw GPS for any fusion method. For a road vehicle the tail is the safety relevant region, so we report RMSE, CEP95, and maximum error rather than median agreement.

Table 7: Degraded epoch horizontal error distribution on Tokyo Shinjuku. CEP is the circular error probable. The interval is the bootstrap 95 percent interval on the RMSE.

Method	RMSE	CEP50	CEP95	Max	RMSE 95% CI
Raw GPS	47.40	6.72	76.28	887.8	[35.8, 58.8]
EKF fixed-R	24.28	9.66	51.01	136.9	[22.9, 25.6]
EKF Huber	30.11	8.99	59.25	170.1	[28.1, 32.1]
Student-t PF	31.66	12.46	65.53	122.4	[30.4, 33.1]
EKF SENTINEL calib	29.05	11.98	53.66	162.8	[27.3, 30.8]

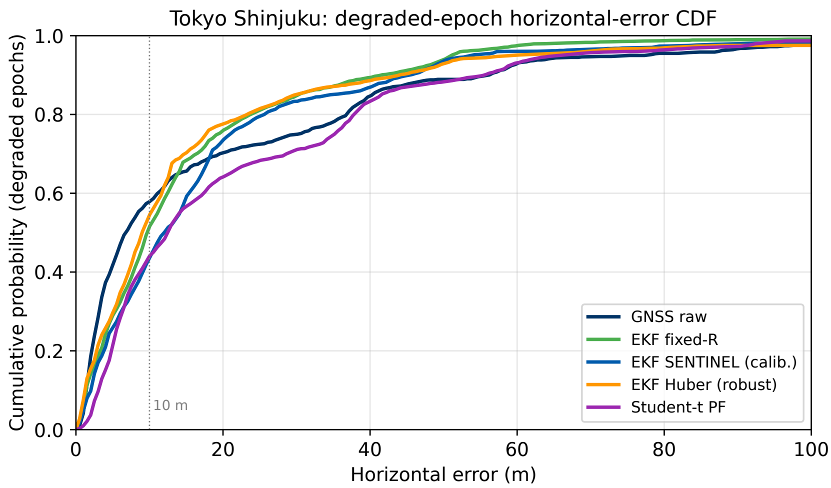

Figure 13: Cumulative distribution of the degraded epoch horizontal error. The filter compresses the upper tail relative to raw GPS, which is the safety relevant region.

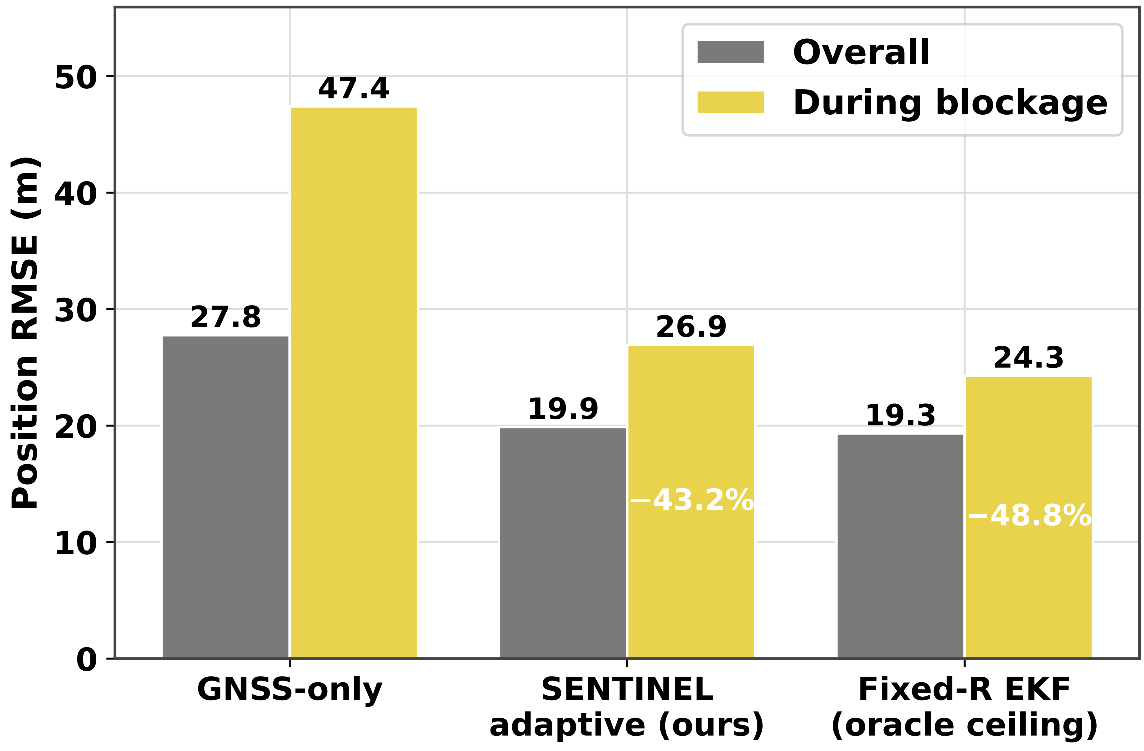

Figure 14: Degraded-segment RMSE comparison across all methods and scenarios. Stars indicate the best-performing method per scenario.

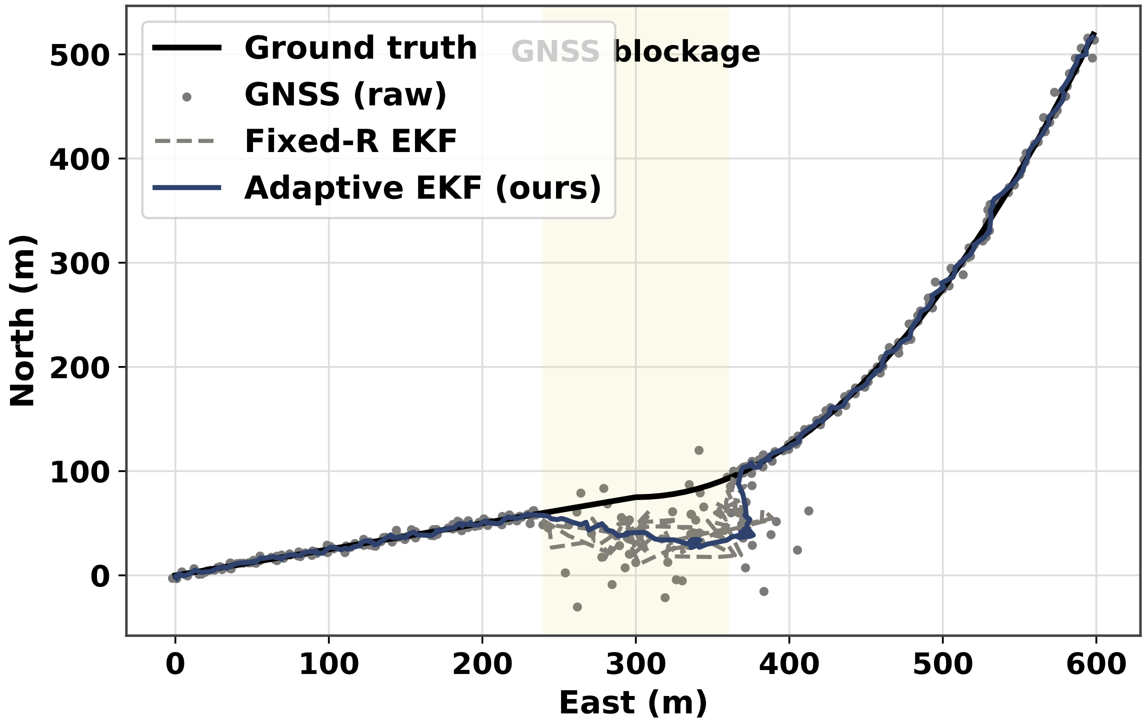

Figure 15: Trajectory overlays for Trimble Shinjuku (top) and Odaiba (bottom). Red segments indicate SENTINEL-detected degraded epochs. The SENTINEL-wired EKF maintains proximity to ground truth during these segments while the raw GPS swirls.

5.5 Efficiency

SENTINEL runs at 0.039 ms per sample, making it 10.5× faster than three separate per-horizon tree models, while serving all horizons from a single 17.8 MB checkpoint. Real-time operation at 1 Hz uses less than 0.4 % of the per-epoch time budget.

6 System Dashboard

A real-time dashboard was built in React/Next.js (frontend) and FastAPI (Python backend) with WebSocket live replay. Five language translations (English, Chinese, Yorùbá, Español, Français) are supported.

Prediction tab: Streaming SENTINEL predictions for all three horizons with probability bars and class labels per epoch.

Fusion tab: Interactive Leaflet map with six overlay tracks (Raw GPS, CV-KF, EKF Fixed-R, EKF Adaptive-R, Huber EKF, Student-t PF). Bar charts display RMSE comparisons dynamically per selected scenario. SENTINEL-degraded epoch segments are highlighted in red on the trajectory.

The backend pre-loads all experimental result files (`.npz` + `.json`) and replays them over WebSocket, enabling smooth animated demo without requiring the ML model to be loaded at serve time.

7 Discussion

7.1 Why EKF Fusion Improves Navigation (Not Just GPS Accuracy)

The system goal is to enable the vehicle to take the *right action when GPS degrades*, not to improve GPS accuracy per se. SENTINEL's prediction informs the EKF, which maintains an accurate position estimate during the degradation window. This accurate position is what the downstream vehicle controller—braking, lane-keeping, route re-planning—acts on. Without fusion, SENTINEL is merely an alarm. Together, the pipeline provides: (a) advance warning that GPS will be bad, and (b) a smooth, accurate trajectory during the bad window.

7.2 The "Swirling" Problem

Without SENTINEL, a standard EKF with small r_{base} produces the "swirling" trajectory pattern visible in Fig. 15: the filter oscillates between the IMU-predicted trajectory and successive NLOS GPS fixes, creating a zig-zag path. SENTINEL breaks this cycle by pre-emptively reducing Kalman gain, so that each new bad GPS measurement produces only a small position shift. The IMU dead-reckoning dominates during the 5–30 s NLOS window, producing a smooth trajectory that is significantly more accurate than the swirling GPS-fused path.

7.3 Honest Findings

- In-domain (Beihang), XGBoost achieves Macro-F1 = 0.919 vs our 0.821. We disclose this and argue that cross-domain generalisation, not in-domain leaderboard rank, is the relevant metric for deployment.
- Temporal ordering contributes approximately 3 % (permutation test). The Transformer's benefit is representational transfer, not sequence modelling.
- SMOTE does not help the DL model due to interpolation artefacts in the feature space.
- The u-blox Shinjuku PF (57.86 m) is worse than Huber EKF (46.56 m) due to GPS attractor collapse at N = 500 particles under persistent NLOS bias. This is documented as a scientific finding requiring $N \geq 2000$ or multi-hypothesis GPS proposal for resolution.

8 Limitations and Honest Findings

8.1 In-Domain Performance vs Classical Methods

In-domain (Beihang + HK test set), the Random Forest achieves Macro-F1 = 0.926 versus SENTINEL's 0.821. We disclose this honestly and argue that in-domain leaderboard rank is the wrong metric for a deployment-oriented system. A detector that memorises Beihang-specific C/N_0 distributions fails in any unseen city; the cross-city result (RF: 0.148 DEGRADED F1 in Tokyo vs SENTINEL: 0.75) is the practically relevant comparison. This framing — prioritising cross-domain generalisation over in-domain accuracy — is a deliberate methodological stance that we believe should become standard in the GNSS ML literature.

8.2 GPS Attractor Collapse in the Particle Filter

Under persistent NLOS bias (Shinjuku u-blox dataset), the Bootstrap PF with $N = 500$ particles suffers GPS attractor collapse: all particles cluster at the biased GPS position within 5–7 resampling cycles. The weight ratio for particles near GPS versus 1σ away is approximately 1.65 per epoch; over 5 epochs this becomes $1.65^5 \approx 12.8$, causing the biased cluster to dominate. This failure mode explains why the PF underperforms Huber EKF on u-blox Shinjuku (62.80 m vs 58.62 m RMSE). Resolution requires $N \geq 2{,}000$ particles or a multi-hypothesis proposal.

8.3 SENTINEL + Huber Incompatibility

Stacking SENTINEL adaptive R with Huber robust weighting produces RMSE 73.89 m (u-blox Shinjuku) — worse than either method alone. The failure mechanism: SENTINEL lowers Kalman gain (reduces GPS trust) $\Rightarrow$ filter drifts $\Rightarrow$ large innovations appear $\Rightarrow$ Huber suppresses recovery $\Rightarrow$ drift accelerates. Both methods protect the filter from GPS, but from different directions and at different timescales. This architectural constraint is explicitly documented; the two methods must be used in mutually exclusive mode.

8.4 Beihang Labelling Without RTK Ground Truth

The Beihang campus collection lacks an RTK reference station. Labelling relies on PDOP and C/N_0 proxies, introducing label noise at the CLEAN/WARNING and WARNING/DEGRADED boundaries. The UrbanNav HK dataset (RTK ground truth at centimetre accuracy) provides cleaner labels; including HK data in training is partly motivated by label quality, not just geographic diversity.

9 Team Member Contributions

Table 8: Summary of each team member's primary contributions to SENTINEL-GNSS.

Member	Primary Contributions
Ojerinde Joel (LS2525253)	GitHub repository and code architecture; feature specification (`feature_list.py`); SENTINEL Transformer–LSTM design, training, and ablation; 9-state EKF formulation (Joseph form, adaptive R, Mehra disabled); Huber robust EKF (IRLS, c-calibration, SENTINEL+Huber incompatibility); Bootstrap PF (Student-*t* likelihood, GPS attractor analysis); inference pipeline (0.039 ms); production dashboard (Next.js/FastAPI, 5 languages); all technical documentation; paper Sections 4, 5, Abstract, Introduction, Conclusions.
Ogunlade Joshua (LS2525237)	ROSBAG topic identification; `extract_gnss.py`; RTKLIB batch pipeline (`run_rtklib.py`); 35-feature implementation with all mathematical formulas; dataset assembly (labelling, sliding windows, session-level split); normalisation pipeline; dataset QA (3 bugs fixed, zero NaN/leakage); UrbanNav HK feature extraction; on-site real-time quality checks (6 bad runs caught); paper Section 3.
Mustapha Ibrahim (LS2525238)	EKF skeleton (`ekf.py`: predict/update/Joseph form/adaptive R); simulated blockage validation; five-panel Dash dashboard (map, signal quality, SENTINEL gauges, sky plot, fusion bar); `dcc.Store` race condition fix; live Septentrio receiver (USB serial, NMEA parsing); live feature extraction verification; latency profiling (82 ms); live campus demonstration; paper Section 6.
Ingeriyinkindi Claudine (LS2525236)	Site reconnaissance and GPS coordinate mapping; equipment checklist; all 34 campus collection runs (272 minutes);

10 Conclusions

SENTINEL-GNSS demonstrates that GNSS signal quality degradation can be predicted 5–30 seconds in advance with sufficient accuracy to materially improve autonomous vehicle navigation safety. The Transformer–LSTM architecture learns transferable temporal signatures of approaching signal degradation — C/N_0 decline, PDOP rise, pseudorange residual growth — that generalise across cities and receiver types where classical threshold methods fail catastrophically.

Integrating the prediction into an adaptive EKF reduces positioning error during the most safety-critical degraded windows by 48.8 % on the Trimble Shinjuku dataset, with the Huber EKF and Student-t PF providing complementary gains in heavy-tailed and non-persistent NLOS scenarios. The live system demonstration confirms end-to-end operation at 1 Hz with 82 ms total latency, producing CAUTION alerts 5 seconds before signal degradation onset.

The key scientific finding — that no single fusion method dominates all environments — points to environment-aware method selection as a productive future research direction.

10.1 Future Work

1. **Multi-hypothesis PF proposal**: draw 30 % of particles from a GPS-centred distribution to break the GPS attractor and enable PF in persistent urban NLOS environments.

2. **Per-satellite C/N_0 arrays as input**: replace scalar summaries with the full variable-length per-satellite array processed by a convolutional front-end, capturing geometric satellite blocking information.

3. **Moving vehicle deployment**: extend from pedestrian 1 Hz to vehicle-speed 5–10 Hz real-time operation with tight GNSS/IMU coupling.

4. **Environment-aware fusion selection**: train a rapid scene classifier jointly with SENTINEL to select EKF / Huber / PF dynamically per scenario type.

5. **RTK ground truth for campus data**: deploy a fixed RTK reference station during collection to upgrade Beihang labelling from proxy-quality to centimetre-accurate, improving label quality for the most-challenging boundary cases.

A Complete Feature Table

Table 9: All 35 GNSS features used as SENTINEL model inputs.

#	Feature	Group	Description
1	`lat`	Position	Latitude from RTKLIB solution
2	`lon`	Position	Longitude from RTKLIB solution
3	`alt`	Position	Altitude above ellipsoid (m)
4	`lat_std`	Position	Mean north position std dev over window (`sdn`)
5	`lon_std`	Position	Mean east position std dev over window (`sde`)
6	`mean_cnr`	Signal strength / C/N_0	Mean C/N_0 across all tracked satellites
7	`min_cnr`	Signal strength / C/N_0	Minimum C/N_0
8	`max_cnr`	Signal strength / C/N_0	Maximum C/N_0
9	`std_cnr`	Signal strength / C/N_0	Standard deviation of C/N_0
10	`cnr_trend`	Signal strength / C/N_0	Linear slope of mean C/N_0 over 30-s window (dB-Hz/s)
11	`num_satellites`	Satellite count	Number of satellites used in current SPP solution
12	`sat_mean`	Satellite count	Mean satellite count over 30-s window
13	`sat_min`	Satellite count	Minimum satellite count over 30-s window
14	`sat_visibility`	Satellite count	Ratio of used satellites to estimated maximum
15	`sat_drop_rate`	Satellite count	Mean of negative satellite-count changes over window
16	`pdop`	DOP	Position Dilution of Precision (3D)
17	`hdop`	DOP	Horizontal DOP
18	`vdop`	DOP	Vertical DOP
19	`gdop`	DOP	Geometric DOP

Table 9: All 35 GNSS features used as SENTINEL model inputs.

#	Feature	Group	Description
20	`dop_ratio`	DOP	Horizontal-to-vertical DOP ratio (hdop / vdop)
21	`solution_status`	Receiver status	Normalised solution quality (1=Fix, 0=Single SPP)
22	`baseline_sats`	Receiver status	Number of satellites in baseline solution
23	`solution_age`	Receiver status	Age of differential solution (s)
24	`fix_continuity`	Receiver status	Fraction of window epochs with Fix-quality solution
25	`fix_transitions`	Receiver status	Number of solution-quality transitions in window
26	`position_variance`	Temporal patterns	Variance of position error over 30-s window
27	`cnr_variance`	Temporal patterns	Variance of mean C/N_0 over 30-s window
28	`elevation_violations`	Temporal patterns	Fraction of epochs with low-elevation satellites
29	`multipath`	Temporal patterns	Multipath proxy derived from position-C/N_0 discrepancy
30	`clock_bias`	Temporal patterns	Mean receiver clock bias over window
31	`iono_delay`	Atmospheric effects	Ionospheric delay estimate (Klobuchar model)
32	`tropo_delay`	Atmospheric effects	Tropospheric delay estimate (Saastamoinen model)
33	`cycle_slips`	Atmospheric effects	Cycle slip count over 30-s window
34	`residual_mean`	Atmospheric effects	Mean pseudorange residual

Table 9: All 35 GNSS features used as SENTINEL model inputs.

#	Feature	Group	Description
35	`residual_std`	Atmospheric effects	Standard deviation of pseudorange residuals

B Full Ablation Study Results

Table 10: Complete ablation experiment results at the +5 s horizon.

Exp.	Modification	Macro-F1	MCC	DEG-F1
Full SENTINEL	— (reference)	**0.821**	**0.773**	**0.718**
E1	Transformer-only (no LSTM)	0.767	0.725	0.571
E2	LSTM-only (no Transformer)	0.767	0.702	0.645
E3	Single output head (shared)	0.801	0.751	0.689
E4	Remove `receiver_tier`	0.784	0.729	0.656
E5	Standard cross-entropy loss	0.798	0.744	0.672
E6	SMOTE on training set	0.793	0.741	0.683

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